CSC4197

RESEARCH METHODOLOGY

AI IN RESEARCH WRITING, ETHICS & ACADEMIC INTEGRITY

2 THE AI TOOLBOX FOR RESEARCHERS (2026)

Stage	Tools
Brainstorming & outlining	Claude, ChatGPT, Gemini
Literature discovery	Elicit, Scite, Consensus, Semantic Scholar, Research Rabbit, SciSpace
Reading assistance	LLM summarization, NotebookLM-style grounded Q&A
Language polishing	Grammarly, LanguageTool, LLM editing passes
Coding & analysis	Copilot/Claude Code, LLM-assisted data analysis
Transcription	Whisper-based tools for interviews

These tools are powerful — and every one of them can be misused.

3 THEORY: THE ACCEPTABLE-USE SPECTRUM

- **Generally acceptable (with disclosure where required):**
  - Grammar/clarity polishing of **your own** ideas and drafts
  - Brainstorming directions, counter-arguments, structure
  - Explaining concepts and code you then verify
  - Searching/summarizing literature **that you then read yourself**
- **Generally unacceptable:**
  - Submitting AI-generated text as your own intellectual contribution
  - **Hallucinated citations** — references that don't exist
  - AI-fabricated data, results, or participant quotes
  - Listing AI as an **author** (all major publishers prohibit this — accountability cannot be delegated)

4 PUBLISHER & INSTITUTIONAL AI POLICIES

- Common ground across IEEE, ACM, Elsevier, Springer (verify current text per venue):
- AI tools **cannot be authors**
- Substantive AI use must be **disclosed** (acknowledgments/methods)
- Authors remain **fully accountable** for accuracy, originality, and integrity of every word
- AI-generated **images** face stricter or total restrictions
- Reviewers generally may **not** upload confidential manuscripts to AI tools
- Your obligation: check the **specific venue policy** + AIUB's academic-integrity policy before submission.

5 THE HALLUCINATION HAZARD

- LLMs generate **plausible** text — including plausible-looking but **nonexistent references**
- Real-world consequences: lawyers sanctioned for fake case citations; papers retracted for fabricated references
- **Demo hands on:** Let's asks an LLM for "5 papers on X with DOIs" → you verify each on Crossref/Scholar → count the fakes
- **Rule:** every citation in your work must be opened, read, and verified by a human author. No exceptions.

6 RESEARCH ETHICS: HUMAN PARTICIPANTS

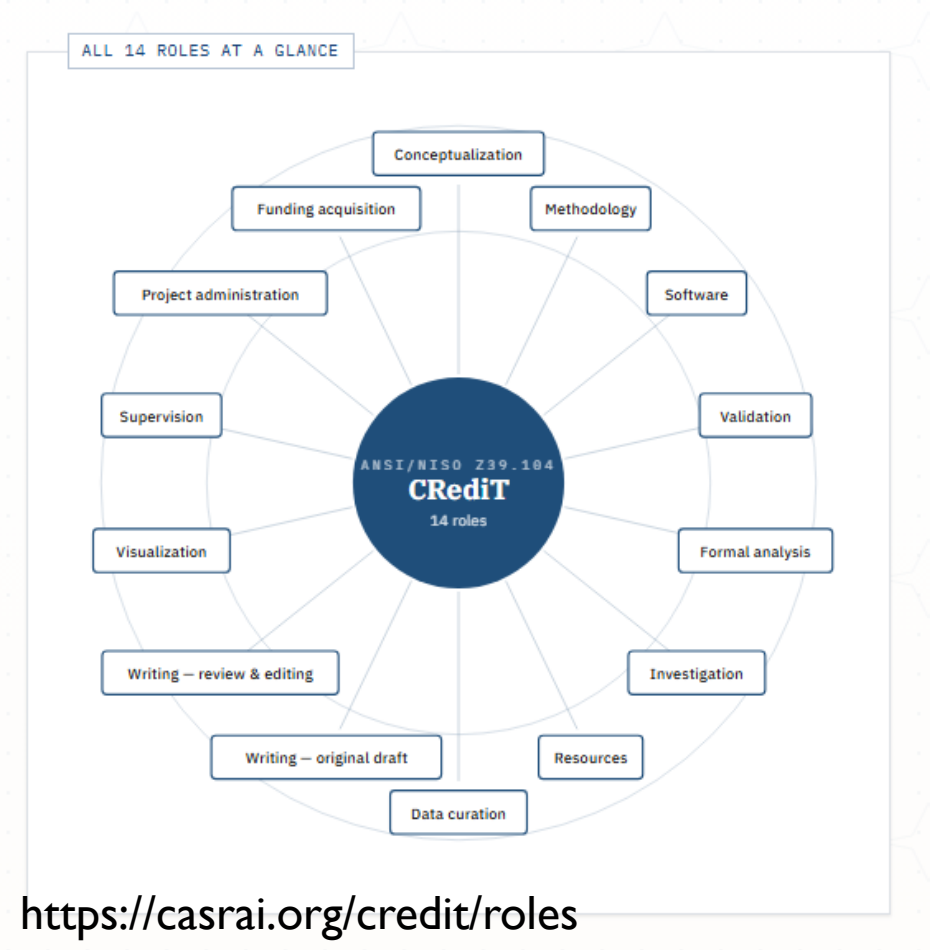
- If your study involves people (surveys, interviews, user studies):
- **Informed consent** — purpose, procedures, risks, right to withdraw, in language participants understand (Bangla version where appropriate)
- **Anonymity** (identity never collected) vs. **confidentiality** (collected but protected) — know the difference, promise only what you can keep
- **Data protection** — secure storage, minimal collection, retention limits, anonymization before sharing
- **Ethical review** — IRB/ethics committee approval where required; minors and vulnerable groups need extra safeguards
- CS-specific: crowdsourcing fair pay, deception in security studies, scraping personal data

ACADEMIC INTEGRITY: THE VIOLATION TAXONOMY

- **Plagiarism** — verbatim copying; **mosaic** (patchwriting with synonyms); **self-plagiarism** (recycling your own submitted work)
- **Fabrication** — inventing data/results
- **Falsification** — manipulating data, images, or omitting inconvenient results
- **Salami slicing** — splitting one study into many minimal papers
- **Ghost/gift authorship** — unlisted contributors / listed non-contributors
- **Contract cheating** — paying someone (or something) to do your work
- Consequences: course failure, degree revocation, retraction, career damage — integrity records are permanent.

8 AUTHORSHIP DONE RIGHT: CREDIT

- **CRediT taxonomy** — 14 standard contributor roles, e.g.: Conceptualization · Methodology · Software · Validation · Formal analysis · Data curation · Writing (original draft) · Writing (review & editing) · Supervision
- Decide authorship & order **early**, document contributions
- Everyone listed must have contributed substantively **and** approve the final version
- Disclose **conflicts of interest** (funding, employment, personal)



https://casrai.org/credit/roles

9 PARAPHRASING PROPERLY + SIMILARITY TOOLS

- **Bad (mosaic):** original sentence with synonyms swapped → still plagiarism
- **Good:** read → close the source → write the idea in your structure → cite
- **Turnitin/iThenticate** similarity reports: % is a *signal*, not a verdict — examine matched sources; quoted+cited matches differ from unattributed ones
- Universities set thresholds (commonly ~15–20% with exclusions) — but 5% unattributed copying is worse than 25% properly-quoted material
- AI-detection tools exist but are **unreliable** (false positives, especially for non-native English writers) — process evidence (drafts, version history) is your best protection

10 CASE STUDIES (DISCUSSION)

- Discuss in groups — verdict + reasoning for each:
- Student uses an LLM to rewrite their methods section for grammar, discloses it in acknowledgments. *Acceptable?*
- Student asks an LLM to write the related-work section, lightly edits, cites the (unverified) references it produced. *What violations?*
- Two course projects share 60% of their text because the same group reused last semester's report. *Which violation? How to do it legitimately?*
- A supervisor adds a colleague as author "for their CV" — the colleague never saw the paper. *Name the practice.*
- Group removes 3 outlier data points that ruin their hypothesis, without reporting it. *Falsification or legitimate cleaning? What would make it legitimate?*

11 REPRODUCIBILITY & OPEN SCIENCE

- The integrity-positive flip side:
- **Share code & data** (GitHub + Zenodo DOI) where licenses/ethics allow
- **Document environment** — seeds, versions, hardware ("Results obtained with seed=42, PyTorch 2.x, single A100")
- **Preregistration** — commit hypotheses/analysis before data collection (kills p-hacking)
- CS venues increasingly run **artifact evaluation** — badges for reproducible papers
- For your project: a README that lets a stranger reproduce your tables = bonus credibility (and viva armor)

12 WRAP-UP & NEXT WEEK

- **Key takeaways**
- AI is a legitimate assistant, never an author — disclose substantive use; verify everything, especially citations
- Ethics = consent + confidentiality + approval; Integrity = your words, your data, your accountability
- Reproducibility turns integrity into a competitive advantage
- **Action item:** add an "AI use & integrity" statement to your project paper acknowledging tools used
- **Next week:** Thesis writing & research dissemination — venues, peer review, rebuttals + in-class peer-review workshop. Bring your full draft progress.